Ph.D. Qualifying Exam 2023: Statistical Inference

July 5, 2023
10 AM to 2 PM

Rules: This is a four hour exam consisting of two problems, both of which should be answered. You may use ONE page (front and back, 8.5 by 11 inch paper) HANDWRITTEN "information sheet" for EACH COURSE (Stat 593 and Stat 594) on which this exam is based. No other references are allowed. No use of cell phones, calculators, computers, or other electronic devices is permitted for any reason during the time while the exam is being administered, including any time during which you leave the exam room for any reason. **Show your work and justify your answers clearly**. **If you use known results (e.g., results stated/proven in class), please state them clearly.** Please put your name or initials on every page. Finally, write your answers clearly and legibly in the space provided after each problem, preferably using pens, leaving an impression deep enough for the papers to be scanned. Good luck!

Problem 1

Instructions:

- If you cannot solve/prove one problem completely, provide as many partial details as possible.

Problem. Consider the linear model

$$Y = X\theta + \varepsilon,$$

where $X \in \mathbb{R}^{n \times p}$ is a design matrix, $\theta \in \mathbb{R}^p$ is an unknown regression vector and $\varepsilon \sim \mathcal{N}(0, I_n)$.

1. Fix $\theta_0, \theta_1 \in \mathbb{R}^p$. Compute the maximal power of any level-α test for the testing problem $H_0 : \theta = \theta_0$ versus $H_1 : \theta = \theta_1$.

2. Suppose we place a multivariate normal prior $\mathcal{N}(0, \Lambda)$ on θ for some non-singular covariance matrix $\Lambda \in \mathbb{R}^{p \times p}$.

   (a) Further assume that the sample covariance $X^\top X/n \to \Sigma$ as $n \to \infty$ with p fixed (note this holds almost surely if the rows of X are i.i.d. $\mathcal{N}(0, \Sigma)$). Prove directly a Bernstein von-Mises theorem for the posterior distribution $\Pi(\theta|Y)$ of θ.

(b) Consider a special form of $\Lambda = \lambda I_p$. Find the posterior mean as $\lambda \to \infty$ when X is full rank.

Problem 2

Suppose that the observed data (x_i, y_i), $1 \le i \le n$, are independent and identically distributed observations as (x, y), where y is a response variable and $x = (x^{(1)}, \ldots, x^{(p)})^{\mathrm{T}}$ is a $p \times 1$ covariate vector, such that each covariate has mean 0, i.e., $E(x^{(j)}) = 0$ for $j = 1, \ldots, p$. The problem of interest is to estimate the mean $\alpha^* = E(y)$, while taking advantage of the fact that $E(x^{(j)}) = 0$ for $j = 1, \ldots, p$. Each covariate $x^{(j)}$ can be called an auxiliary variable or a control variate.

For each question, you may use the results from the preceding questions, whether or not you have answered those questions.

(i) Let $(\hat{\alpha}, \hat{\beta})$ be the least-squares estimator defined as

$$(\hat{\alpha}, \hat{\beta}) = \mathrm{argmin}_{\alpha \in \mathbb{R}, \beta \in \mathbb{R}^p} \ell_n(\alpha, \beta)$$

where $\ell_n(\alpha, \beta) = \frac{1}{2n} \sum_{i=1}^n (y_i - \alpha - x_i^{\mathrm{T}} \beta)^2$. Show that

$$\hat{\alpha} = \bar{y} - \bar{x}^{\mathrm{T}} \hat{\beta},$$

where $\bar{y} = n^{-1} \sum_{i=1}^n y_i$ and $\bar{x} = n^{-1} \sum_{i=1}^n x_i$.

(ii) Assume that $\Sigma = E(xx^{\mathrm{T}})$ is non-singular. As $n \to \infty$ and p fixed, show that $\sqrt{n}(\hat{\alpha} - \alpha^*)$ converges in distribution to $\mathrm{N}(0, V)$, with variance matrix V. Find $\varphi(x, y)$ such that $V = E\{\varphi^2(x, y)\}$.

For the remaining questions, p may be greater than n as n tends to ∞.

(iii) Let $(\tilde{\alpha}, \tilde{\beta})$ be a Lasso estimator defined as

$$(\tilde{\alpha}, \tilde{\beta}) = \operatorname{argmin}_{\alpha \in \mathbb{R}, \beta \in \mathbb{R}^p} \left\{ \ell_n(\alpha, \beta) + \lambda \|\beta\|_1 \right\}, \tag{1}$$

where $\|\beta\|_1$ is the L_1 norm of β and $\lambda > 0$ is a (pre-specified) tuning parameter. Show that

$$\tilde{\alpha} = \bar{y} - \bar{x}^{\mathrm{T}} \tilde{\beta}.$$

(iv) Assume that each covariate is bounded in absolute values by a constant $B_0 > 0$, i.e, $|x^{(j)}| \leq B_0$ almost surely for $j = 1, \ldots, p$. Under some further technical conditions, $(\tilde{\alpha}, \tilde{\beta})$ is expected to converge to (α^*, β^*) in the following sense, with $(\alpha^*, \beta^*) = \mathrm{argmin}_{\alpha \in \mathbb{R}, \beta \in \mathbb{R}^p} E\{(y - \alpha - x^{\mathrm{T}}\beta)^2\}$:

$$\frac{1}{n}\sum_{i=1}^{n}(\tilde{\alpha} - \alpha^* + x_i^{\mathrm{T}}(\tilde{\beta} - \beta^*))^2 = (1 + |S_{\beta^*}|)\lambda_0^2 O_p(1), \tag{2}$$

$$|\tilde{\alpha} - \alpha^*| + \|\tilde{\beta} - \beta^*\|_1 = (1 + |S_{\beta^*}|)\lambda_0 O_p(1), \tag{3}$$

where $\lambda_0 = \sqrt{\log(1+p)/n}$, and $|S_{\beta^*}|$ is the size of the set $S_{\beta^*} = \{1 \leq j \leq p : \beta_j^* \neq 0\}$ with $\beta^* = (\beta_1^*, \ldots, \beta_p^*)^{\mathrm{T}}$. Please formulate a set of technical conditions (as weak as you can) under which you expect that (2) and (3) can be established. You *do not* need to prove (2) and (3) under your conditions.

(v) As one of the steps to prove (2) and (3), you are *only* asked to prove the following result. For $\delta > 0$, formulate an event Ω_0 such that $P(\Omega_0) \geq 1 - 2\delta$ and the following inequality holds under event Ω_0 with $\lambda = A_0 \lambda_{0,\delta}$ in (1) for a constant $A_0 > 1$:

$$
\begin{aligned}
&\frac{1}{n} \sum_{i=1}^{n} (\tilde{\alpha} - \alpha^* + x_i^{\mathrm{T}} (\tilde{\beta} - \beta^*))^2 + (A_0 - 1)\lambda_{0,\delta} (|\tilde{\alpha} - \alpha^*| + \|\tilde{\beta} - \beta^*\|_1) \\
&\quad \leq 2 A_0 \lambda_{0,\delta} \big(|\tilde{\alpha} - \alpha^*| + \sum_{j \in S_{\beta^*}} |\hat{\beta}_j - \beta_j^*| \big),
\end{aligned}
$$

where $\lambda_{0,\delta} = C_0 \sqrt{\frac{log((1+p)/\delta)}{n}}$ for some constant C_0. State your C_0 explicitly.

(vi) Suppose that (2) and (3) hold. Show that $\sqrt{n}(\tilde{\alpha} - \alpha^*)$ converges in distribution to $\mathrm{N}(0, V)$, with variance matrix V as in Question (ii), provided $(1 + |S_{\beta^*}|) \log(1 + p) = o(n^{1/2})$.